??????????? ?????? ?2

?????????????? ? ?????????? ????????? ??????????

??????? 7: ?????????? ?????????? ??????? (???? 3?3 ? 25?25).

? ?????? ?????????? ??? ???????? ??? ???????????? ??????????? ??????? ???????? ? ??????? ? ???????????:
1) ?????????? ???????????? (RGB) ?????????? ? ???????????? (BMP).
2) ?????????? ??????????? ?????????? ? ??????????? ??????? ???????.

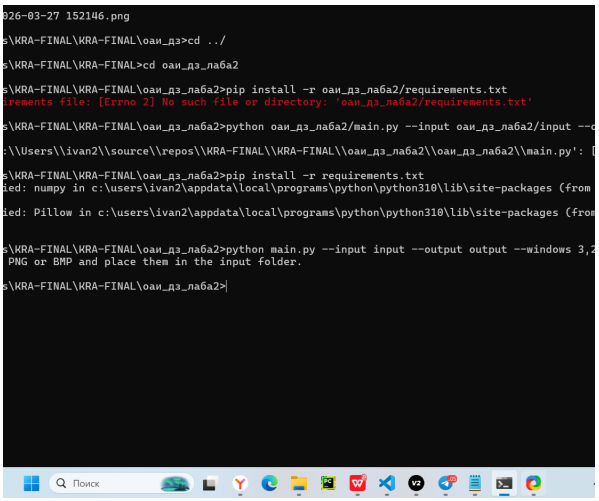
1) ?????????? ? ????????

??????? ??????? ?????????? ?????????? ?????????? ???????:
Y = 0.299R + 0.587G + 0.114B
????????? ?????????? ??? 8-bit grayscale BMP (*_gray.bmp).

2) ?????????? (???????)

????????? ????? ??? ??????? (x, y): T(x,y) = m(x,y) * (1 + k * (s(x,y)/R - 1)).
?????????: k = 0.5, R = 128.
???? Y(x,y) > T(x,y) ? 255, ????? ? 0. ??????????????? ???? 3?3 ? 25?25.

??????????



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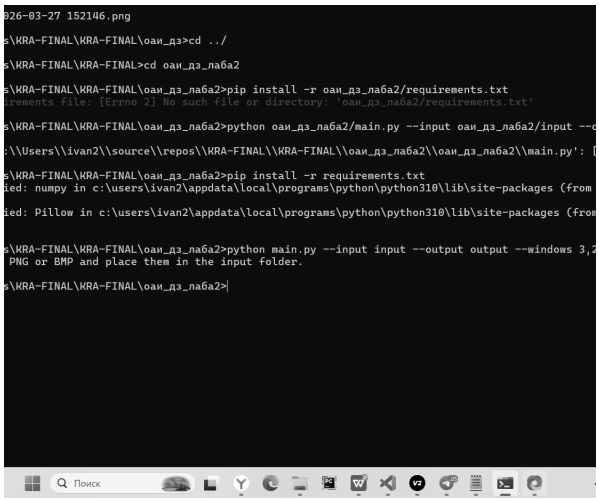

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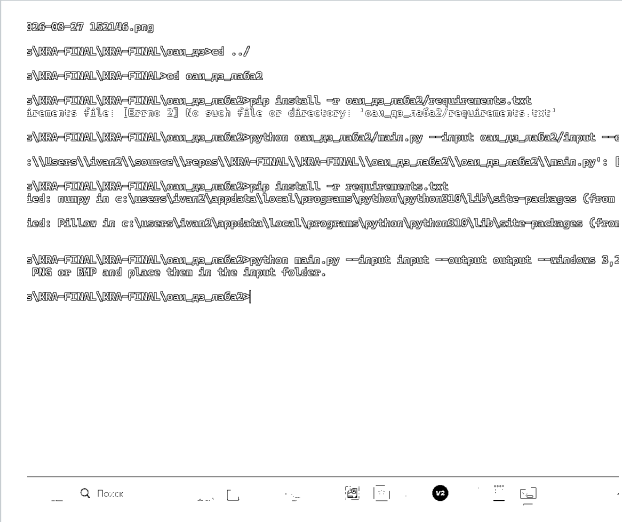
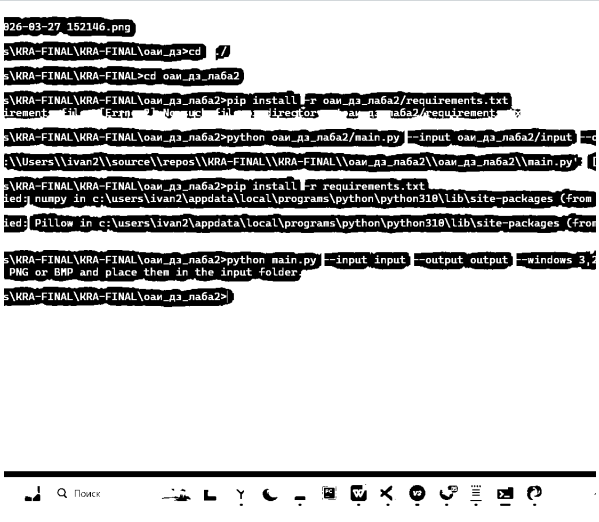

026-03-27 152146.png
s\KRA-FINAL\KRA-FINAL\оаи_дз>cd ../
s\KRA-FINAL\KRA-FINAL>cd оаи_дз_лаба2
s\KRA-FINAL\KRA-FINAL\оаи_дз_лаба2>pip install -r оаи_дз_лаба2/requirements.txt
irements.txt: [Errno 2] No such file or directory: 'оаи_дз_лаба2/requirements.txt'
s\KRA-FINAL\KRA-FINAL\оаи_дз_лаба2>python оаи_дз_лаба2/main.py --input оаи_дз_лаба2/input
:\Users\ivan2\source\repos\KRA-FINAL\KRA-FINAL\оаи_дз_лаба2\оаи_дз_лаба2\main.py':
s\KRA-FINAL\KRA-FINAL\оаи_дз_лаба2>pip install -r requirements.txt
ied: numpy in c:\users\ivan2\appdata\local\programs\python\python310\lib\site-packages (from
ied: Pillow in c:\users\ivan2\appdata\local\programs\python\python310\lib\site-packages (from
s\KRA-FINAL\KRA-FINAL\оаи_дз_лаба2>python main.py --input input --output output --windows 3,
PNG or BMP and place them in the input folder.
s\KRA-FINAL\KRA-FINAL\оаи_дз_лаба2>

??????? 3?3


026-03-27 152146.png
s\KRA-FINAL\KRA-FINAL\оаи_дз>cd ../
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s\KRA-FINAL\KRA-FINAL\оаи_дз_лаба2>pip install -r оаи_дз_лаба2/requirements.txt
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s\KRA-FINAL\KRA-FINAL\оаи_дз_лаба2>pip install -r requirements.txt
ied: numpy in c:\users\ivan2\appdata\local\programs\python\python310\lib\site-packages (from
ied: Pillow in c:\users\ivan2\appdata\local\programs\python\python310\lib\site-packages (from
s\KRA-FINAL\KRA-FINAL\оаи_дз_лаба2>python main.py --input input --output output --windows 3,
PNG or BMP and place them in the input folder.
s\KRA-FINAL\KRA-FINAL\оаи_дз_лаба2>

??????? 25?25